

21.1 Properties and Uses of Alternating Current

Question Paper

Course	CIEA Level Physics
Section	21. Alternating Currents
Topic	21.1 Properties and Uses of Alternating Current
Difficulty	Easy

Time allowed: 40

Score: /31

Percentage: /100

Question 1a

For an alternating voltage, state what is meant by

- (i)
the peak voltage, [1]
- (ii)
the root mean square voltage. [2]

[3 marks]**Question 1b**

A generator produces an alternating voltage which can be described by the equation

$$V = 150 \sin(200\pi t)$$

where V is measured in volts and t is in seconds.

For this alternating voltage, determine

- (i)
the peak voltage, [1]
- (ii)
the r.m.s. voltage, [1]
- (iii)
the frequency. [1]

[3 marks]

Question 1c

The alternating voltage is supplied across a $100\ \Omega$ resistor.

Calculate

- (i)
the peak current and hence the r.m.s. current in the resistor, [2]
- (ii)
the mean power dissipated in the resistor, [2]
- (iii)
the peak power dissipated in the resistor. [2]

[6 marks]

Question 1d

State the effect on the output voltage if the frequency of the generator is increased.

[1 mark]

Question 2a

Fig. 1.1a shows an alternating current generator with a rectangular coil rotating at a constant frequency in a uniform magnetic field.

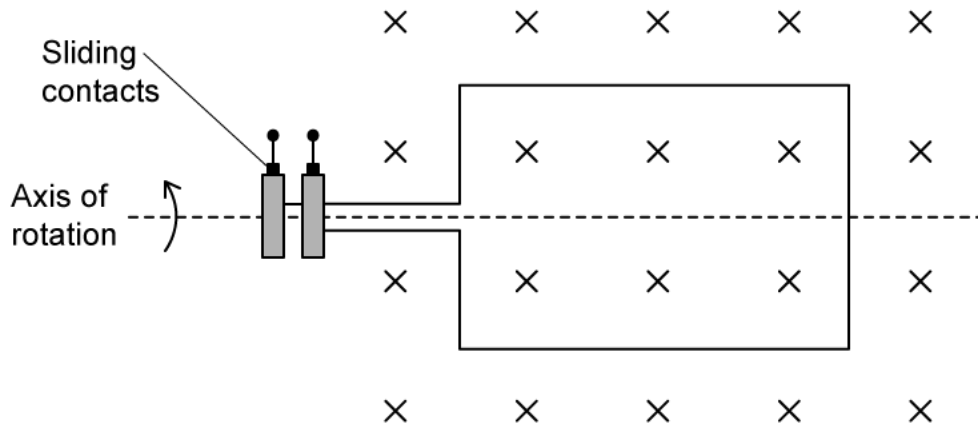


Fig. 1.1a

The graph in Fig. 1.1b shows how the output voltage V from the generator varies with time t .

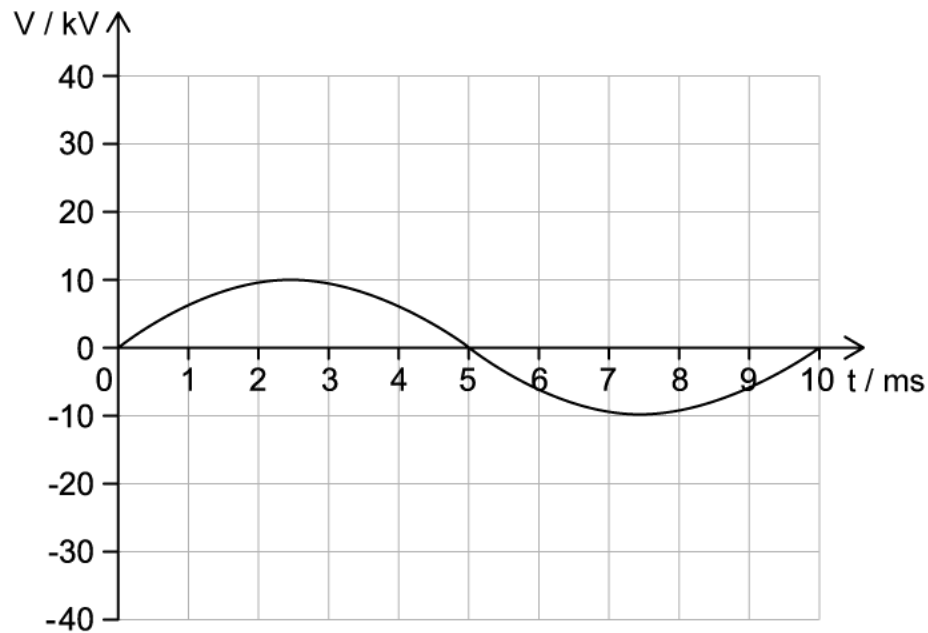


Fig. 1.1b

Using Fig. 1.1b, state the peak output voltage V_0 of the generator.

[1 mark]

Question 2b

Calculate the root mean squared voltage V_{rms} .

[2 marks]**Question 2c**

The mean power output of the generator is 35 kW.

Calculate the root mean squared current I_{rms} .

[3 marks]**Question 2d**

Draw a line on the graph in Fig. 1.1b to show the V_{rms} .

[2 marks]

Question 3a

In rectification, a smoothing capacitor is often necessary.

State the meaning of

(i)
rectification,

[1]

(ii)
smoothing.

[1]

[2 marks]

Question 3b

Fig. 1.1a shows the voltage output from an alternating current supply.

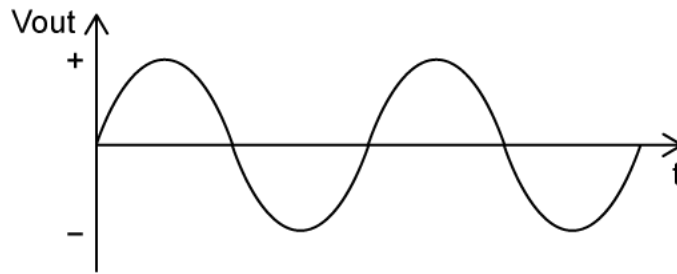


Fig. 1.1a

Sketch the variation of time with output voltage during

(i)

half-wave rectification on Fig. 1.1b



Fig. 1.1b

[2]

(ii)

full-wave rectification on Fig. 1.1c



Fig. 1.1c

[2]

[4 marks]

Question 3c

A capacitor is placed in parallel with a resistive load to smooth the rectified voltage. The graph of the smoothed output voltage against time gives a 'ripple' shape as shown in Fig. 1.2.



Fig. 1.2

State how the 'ripples' in the graph can be reduced.

[2 marks]

Question 3d

Fig. 1.3a shows a diode bridge circuit. It is designed to allow current to flow in certain pathways depending on the input direction of the current.

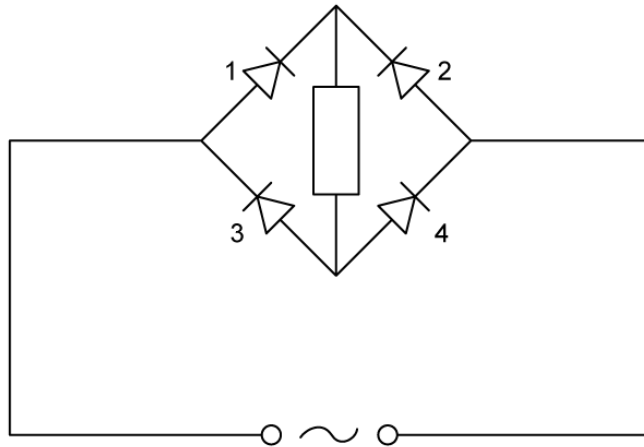


Fig. 1.3a

Draw the path of the current

- (i)
on Fig. 1.3b

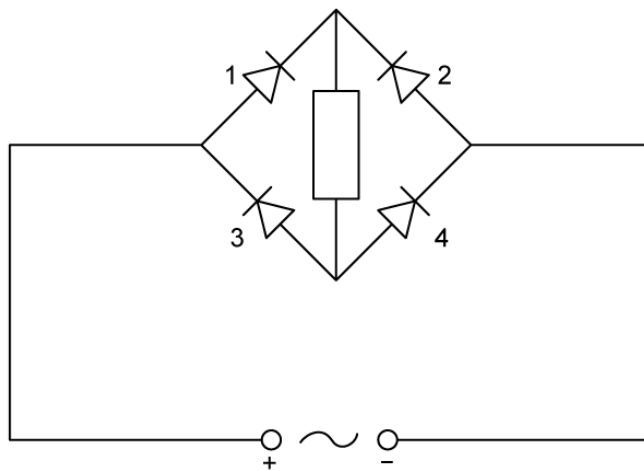


Fig. 1.3b

[1]

- (ii)
on Fig. 1.3c

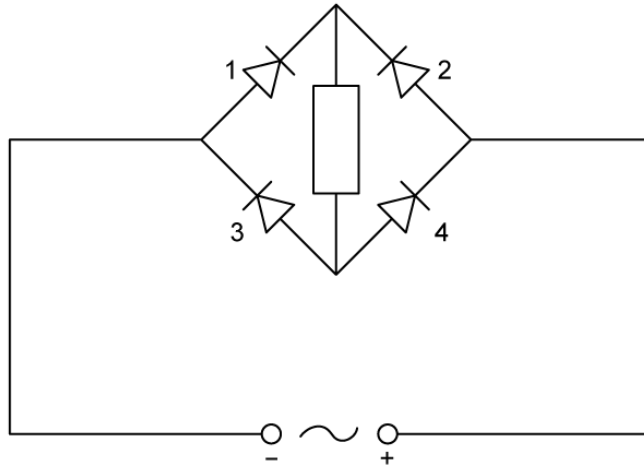


Fig. 1.3c

[1]

[2 marks]



Model Answers: Easy

1a

(a)

(i) Peak voltage is defined as:

- The maximum value of the alternating voltage; [1 mark]

(ii) Root mean square voltage is defined as:

EITHER

- The value of steady / constant voltage; [1 mark]
- That produces same power (in a resistor) as the alternating voltage; [1 mark]

OR

- If the alternating voltage is squared and averaged; [1 mark]
- The r.m.s. value is the square root of this averaged value; [1 mark]

[Total: 3 marks]

1b

(b)

List the known quantities:

- Alternating voltage: $V = 150 \sin(200\pi t)$

(i) Determine the peak voltage:

- Alternating voltage: $V = V_0 \sin \omega t$
- Hence, peak voltage, $V_0 = 150 \text{ V}$ [1 mark]

(ii) Determine the r.m.s. voltage:

- r.m.s. voltage: $V_{rms} = \frac{V_0}{\sqrt{2}}$
- r.m.s. voltage: $V_{rms} = \frac{150}{\sqrt{2}} = 106 \text{ V}$ [1 mark]

(iii) Determine the frequency:

- From $V = V_0 \sin \omega t$: $\omega = 100\pi$
- Frequency: $\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi}$
- Frequency: $f = \frac{200\pi}{2\pi} = 100 \text{ Hz}$ [1 mark]

[Total: 3 marks]

1c

(c)

List the known quantities:

- Peak voltage, $V_0 = 150 \text{ V}$ (from **(b)**)
- r.m.s. voltage, $V_{rms} = 106 \text{ V}$ (from **(b)**)
- Resistance, $R = 100 \Omega$

(i) Calculate the peak and r.m.s. current in the resistor:

- Peak current: $V_0 = I_0 R \Rightarrow I_0 = \frac{V_0}{R}$
- Peak current: $I_0 = \frac{150}{100} = 1.5 \text{ A}$ [1 mark]

- r.m.s. current: $I_{rms} = \frac{I_0}{\sqrt{2}}$
- r.m.s. current: $I_{rms} = \frac{1.5}{\sqrt{2}} = 1.06 \text{ A}$ [1 mark]

(ii) Calculate the mean power dissipated in the resistor

- Mean power: $\langle P \rangle = V_{rms} I_{rms}$
- Mean power: $\langle P \rangle = 106 \times 1.06$ [1 mark]
- Mean power: $\langle P \rangle = 112 \text{ W}$ [1 mark]

(iii) Calculate the peak power dissipated in the resistor:

- Peak power: $P_0 = 2 \times \langle P \rangle$
- Peak power: $P_0 = 2 \times 112$ [1 mark]
- Peak power: $P_0 = 224 \text{ W}$ [1 mark]

[Total: 6 marks]

1d

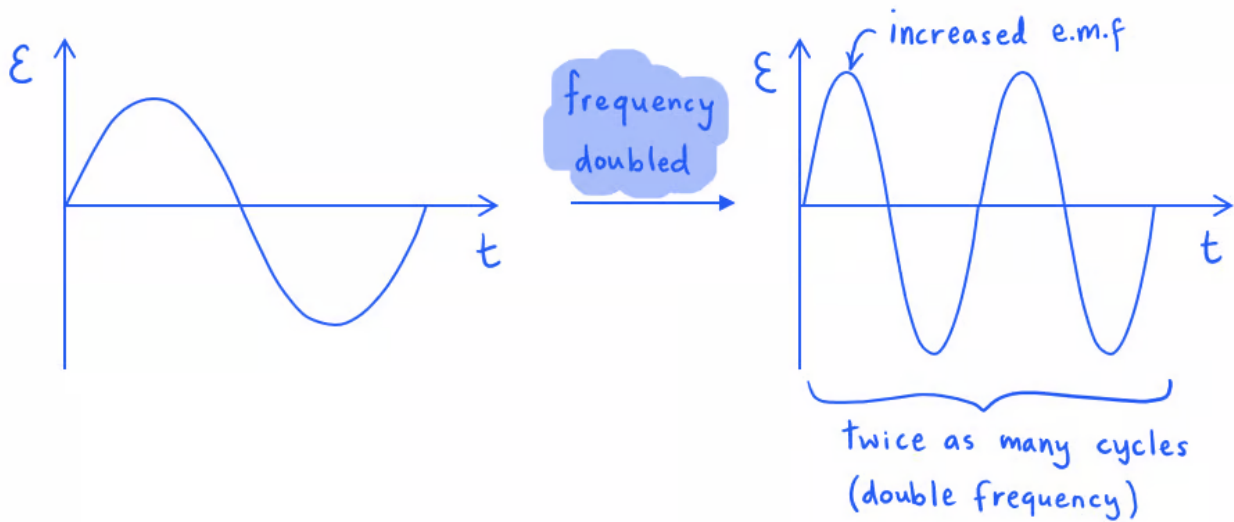
(d) If the frequency of the generator is increased:

- The output voltage will increase; [1 mark]

[Total: 1 mark]

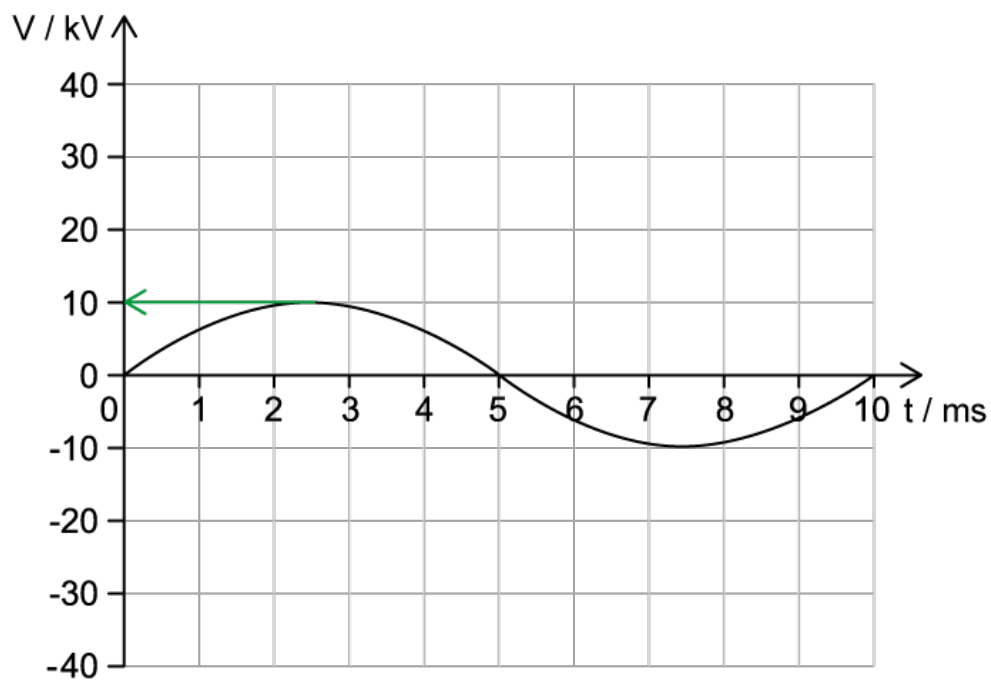
If the frequency of the generator is increased, this means the coil is spinning **faster**. This means the coil will take a shorter time to complete one cycle.

Since the emf generated is the **rate of change** of flux linkage, the **change** in flux linkage will increase. Therefore, the emf increases.



2a

(a) Determine the maximum output voltage of the generator:



- Peak voltage: $V_0 = 10 \text{ kV}$ [1 mark]

[Total: 1 mark]

2b

(b) Calculate the root mean squared voltage:

List the known quantities:

- Peak voltage, $V_0 = 10 \text{ kV} = 10 \times 10^3 \text{ V}$ (from **(a)**)

Use the r.m.s. voltage equation:

- r.m.s. voltage: $V_{rms} = \frac{V_0}{\sqrt{2}}$

- $V_{rms} = \frac{10 \times 10^3}{\sqrt{2}}$ [1 mark]

- $V_{rms} = 7071 = 7.1 \text{ kV}$ [1 mark]

2c

(c) Calculate the root mean squared current:

List the known quantities:

- Average power output, $P = 35 \text{ kW} = 35 \times 10^3 \text{ W}$
- r.m.s. voltage, $V_{rms} = 7.1 \times 10^3 \text{ V}$ (from **(b)**)

Use the equation for mean power:

- Mean power: $P_{mean} = I_{rms} V_{rms} \Rightarrow I_{rms} = \frac{P_{mean}}{V_{rms}}$ [1 mark]

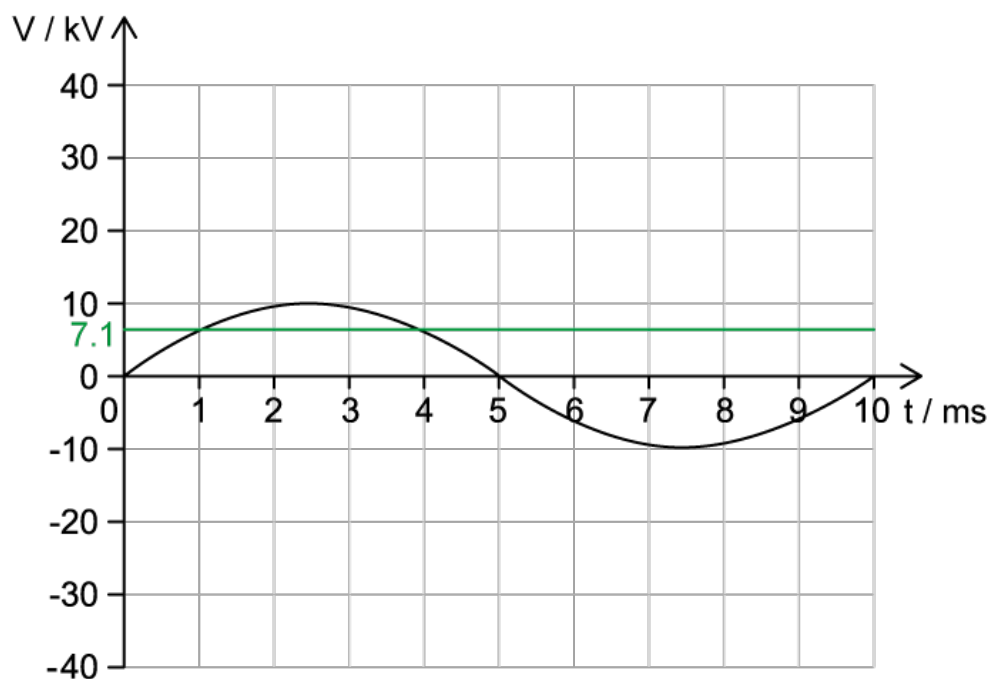
- $I_{rms} = \frac{35 \times 10^3}{7.1 \times 10^3}$ [1 mark]

- $I_{rms} = 4.93 = 4.9 \text{ A}$ [1 mark]

[Total: 3 marks]

2d

(d) Draw a line on the graph in Fig. 1.1b to show the V_{rms} :



- Line draw at 7.1 kV (by eye); [1 mark]
- V-axis labelled; [1 mark]

[Total: 2 marks]

Remember to always draw straight lines with a ruler or straight edge, messy sketches are unlikely to get you full marks in your exam!

3a

(a)

(i) Rectification is defined as:

- The process of converting alternating current (a.c) into direct current (d.c); [1 mark]

(ii) Smoothing is defined as:

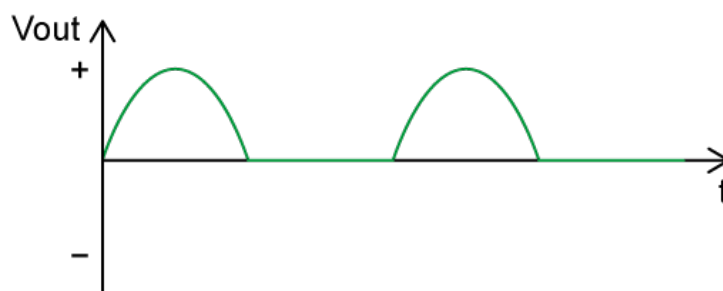
- The reduction in the variation of the output current (or voltage); [1 mark]

[Total: 2 marks]

3b

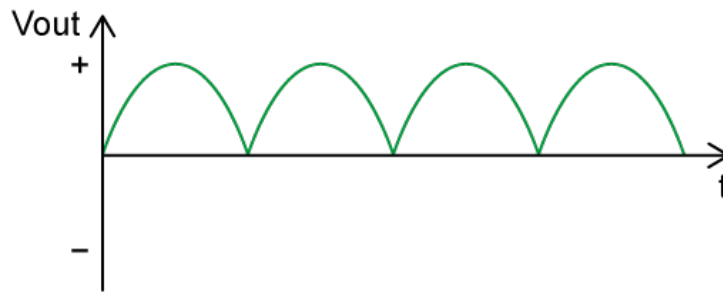
(b) The variation of time with output voltage for...

(i) Half-wave rectification:



- All peaks at the same height; [1 mark]
- Only half a wavelength in the positive direction and zero the rest of the time; [1 mark]

(ii) Full-wave rectification:



- All peaks at the same height; [1 mark]
- All wavelengths in the positive direction; [1 mark]

[Total: 4 marks]

3c

(c) The 'ripples' in the graph can be reduced by:

- A capacitor with greater capacitance; [1 mark]
- A (load) resistor with a larger resistance; [1 mark]

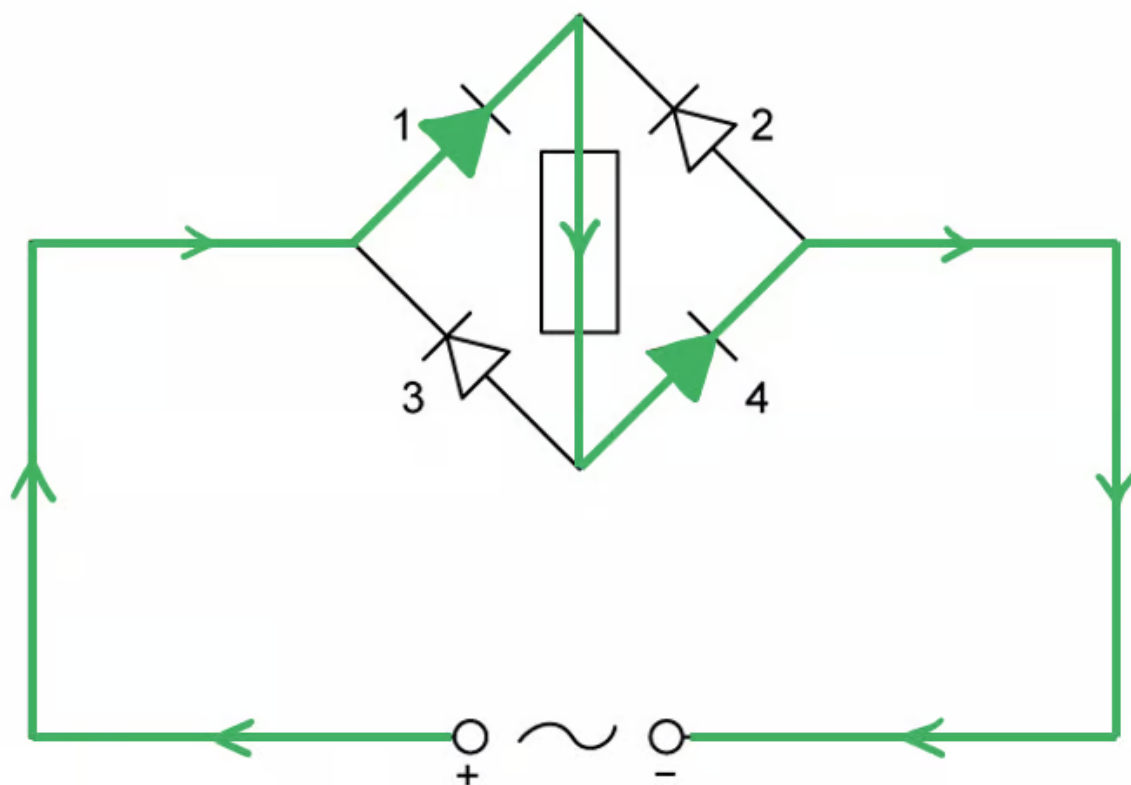
[Total: 2 marks]

This mark scheme requires **two** answers for full marks. Although this is not necessarily stated in the question, it is indicated by the number of marks. A good exam technique is to check the number of marks in the question and make sure you have in your mark scheme to justify them!

3d

(d) The pathways of the current are...

(i) On Fig. 1.3b:



- Current passes through diodes 1 and 4 only; [1 mark]

(ii) On Fig. 1.3c:

